UNIVERSITY OF FLORIDA
Cooperative Extension Service
Institute of Food and Agricultural Sciences

Circular 1143
October 1994

Tomato Yellow Leaf Curl Virus[1]

J. E. Polston, R. J. McGovern, P. A. Stansly[2]

Tomato yellow leaf curl virus (TYLCV) was first characterized in Israel in 1959 and has been a major limiting factor for tomato production in that country since the 1930s. Related viruses have been reported from other Mediterranean countries, Africa, Asia, and Australia. Recently a virus essentially identical to TYLCV from the eastern Mediterranean was found infecting tomato plants in the Dominican Republic (1993) and Jamaica (1994). This virus has drastically limited tomato production in both countries and poses a significant economic threat to the commodity throughout the Western Hemisphere.

TYLCV is a member of the Geminiviridae group of viruses which also contains tomato mottle virus and bean golden mosaic virus which have recently become established in Florida. Like both these viruses, TYLCV consists of twinned (geminate) particles (Figure 1) and is spread by adult silverleaf whiteflies (*Bemisia argentifolii* Perring & Bellows). However, TYLCV consists of a single strand of DNA unlike most other geminiviruses which have two. Analyses of several isolates of TYLCV from various parts of the world have indicated that different viruses can cause the tomato yellow leaf curl disease in different areas of the world. At this time the term TYLCV refers to those virus isolates which are genetically very similar to the virus causing yellow leaf curl first identified in Israel.

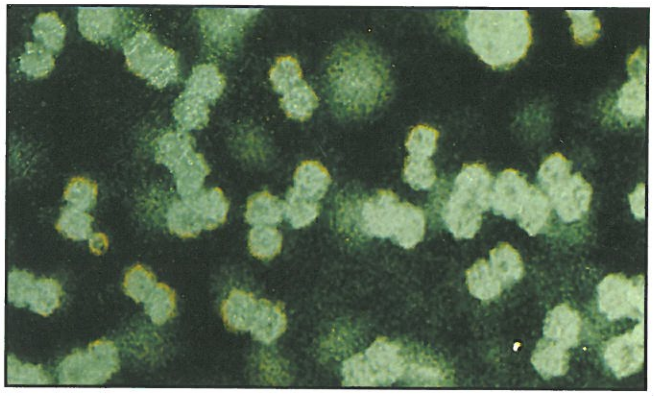


Figure 1. Electron micrograph of geminivirus particles showing the twinned (geminate) shape of the virus particles.

SYMPTOMATOLOGY AND HOST RANGE

Tomato yellow leaf curl virus infects several crop plants but has been reported as a problem only in tomato. The incidence of TYLCV in tomato is often very high and tomato plants are severely affected by the virus. Symptoms vary somewhat among tomato cultivars but in general leaves produced shortly after infection are reduced in size, distorted, cupped inward or downward, and have a yellow mottle. Subsequent leaves become progressively smaller and more distorted, and exhibit both marginal yellowing which sometimes encompasses the entire leaf, and upward leaf cupping (Figures 2 and 3). Infected plants are severely stunted; the younger the plant is at the time

1. This document is Circular 1143, Florida Cooperative Extension Service, Institute of Food and Agricultural Sciences, University of Florida. Publication date: October 1994.

2. J.E. Polston, assistant professor, Plant Virology, GCREC, Bradenton, FL; R.J. McGovern, assistant professor, Plant Pathology, and P.A. Stansly, associate professor, Entomology; SWFREC, Immokalee, FL; Cooperative Extension Service, Institute of Food and Agricultural Sciences, University of Florida, Gainesville FL 32611.

The Institute of Food and Agricultural Sciences is an equal opportunity/affirmative action employer authorized to provide research, educational information and other services only to individuals and institutions that function without regard to race, color, sex, age, handicap, or national origin. For information on obtaining other extension publications, contact your county Cooperative Extension Service office.
Florida Cooperative Extension Service / Institute of Food and Agricultural Sciences / University of Florida / John T. Woeste, Dean



Figure 2. Fresh market tomato cultivar infected with TYLCV. Note upward leaf curling and yellow mottling of leaves.



Figure 3. Processing tomato cultivar infected with TYLCV. Note upward leaf cupping, reduced leaf size, marginal chlorosis and distortion of leaves.

of infection, the more severe the stunting becomes. The most significant effect of TYLCV infection is flower abscission. Fewer than one in ten flowers will set fruit after TYLCV infection, severely reducing yields. The stunting and flower abscission caused by TYLCV are more severe than those caused by tomato mottle virus. Tomato plants infected with both tomato mottle virus and tobacco etch virus have foliar symptoms similar to those of TYLCV, however, without the high frequency of flower abscission. Foliar symptoms of TYLCV lack the purpling of leaf veins and the curling of shoots caused by pseudocurly top virus, a treehopper-transmitted geminivirus found occasionally in tomato in Florida.

Tomato yellow leaf curl virus has a broad host range which currently includes wild and cultivated plants from seven families (Table 1). This contrasts notably with the narrow host range of tomato mottle virus (Fabaceae and Solanaceae). The majority of the host plants infected by TYLCV showed no symptoms, although the virus could be acquired from them and transmitted to tomato by whiteflies. Four species of crop plants (lentil, common bean, tobacco and tomato) have been infected experimentally but only tomato has been found infected naturally. Surveys conducted in Israel showed that only two weed species, *Cynanchum acutum* and cheeseweed (*Malva parviflora*), were infected naturally. In Turkey, jimson weed (*Datura stramonium*), black nightshade (*Solanum nigrum*) and nettle (*Urtica sp.*) were found to be reservoirs of the virus. Table 1 lists the known hosts of TYLCV and whether symptoms occur.

WHITEFLY VECTOR

The sweetpotato whitefly, *Bemisia tabaci* Genn., has recently been split into two species, one named *B. tabaci* (previously referred to as biotype A) and the other named *Bemisia argentifolii* the silverleaf whitefly (previously known as *B. tabaci* biotype B). The two whitefly species are physically difficult to distinguish, but the two species show several biological differences. The silverleaf whitefly feeds and reproduces on a broader range of plant species than the sweetpotato whitefly, including most types of vegetables, broadleaf row crops, weeds, and many ornamentals including poinsettia. The silverleaf whitefly also has a higher rate of reproduction and feeds more heavily than the sweetpotato whitefly. The silverleaf whitefly owes its name to the reaction its feeding produces on squash. Salivary secretions injected into the plant by the immature whitefly produce a silvering of squash leaves, irregular ripening of tomato fruit, and blanching of broccoli stems and poinsettia leaves. In recent years the silverleaf whitefly has become the dominant of the two species throughout the southern United States and in many other areas. The silverleaf whitefly is known to occur on tomatoes in the Dominican Republic and Jamaica and is responsible for the transmission of TYLCV in those areas.

Table 1. Plant Hosts of Tomato Yellow Leaf Curl Virus.

Plant Family	Plant Host	Common Name	Symptoms
Apiaceae	*Chaerogphyllum sp.*	-	No
Asclepiadaceae	*Cynanchum acutum*	-	No
Asteraceae	*Sonchus oleraceae*	Sowthistle	No
Fabaceae	*Lens esculenta*	Lentil	No
	Phaseolus vulgaris	Common bean	No
Malvaceae	*Malva nicaensis*	-	No
	Malva parviflora	Cheeseweed	No
Solanaceae	*Datura stramonium*	Jimson weed	Yes
	Lycopersicon esculentum	Tomato	Yes
	Hyoscyamus desertorum	Henbane	Yes
	Nicotiana benthamiana	-	Yes
	Nicotiana glutinosa	-	Yes
	Nicotiana tabacum	Tobacco	No
	Solanum nigrum	Black nightshade	N/A
Urticaceae	*Urtica sp.*	Nettle	N/A

N/A=information not available.

Adult silverleaf whiteflies are small (1/16 inch in length), winged insects covered with a white powdery wax (Figure 4). They usually alight on the upper leaf surface but quickly move to the lower surface where they feed on plant sap, mate, and lay up to 300 eggs. A tiny "crawler" emerges from each egg and moves across the leaf in search of uninhabited surface. Once settled, the crawler inserts its tube-like stylet and begins to feed. Three molts follow giving rise to successively larger but immobile scale-like stages, the last of which transforms into a yellowish, red-eyed immature ("pupa") approximately 1/32 inch in length. A few days later an adult whitefly emerges. The entire process from egg to adult is about two to four weeks long, depending on the temperatures present during development. Higher temperatures decrease the time required for whitefly development. Once emerged from the "pupa", adult whiteflies remain on the leaf for several hours as they coat themselves with wax which they produce. The whitefly may remain on the plant or fly. Adult whiteflies can fly for several hours and with wind assistance can traverse significant distances.



Figure 4. Adult *Bemisia argentifolii*, the silverleaf whitefly. (Photo by Norm Thomas)

METHODS OF SPREAD

Tomato yellow leaf curl virus is spread primarily by whiteflies. In Israel and the Caribbean the whitefly vector of TYLCV is the silverleaf whitefly. Studies in Israel have shown that the minimum time required for the whitefly to acquire the virus is 15 to 30 minutes, and longer feeding periods result in higher rates of transmission. Similar time periods are required for transmission. Tomato yellow leaf curl virus can be acquired by immature whiteflies but only adults can transmit the virus. Adults do not transmit the virus to their progeny. Tomato yellow leaf curl virus is transmitted by adults for approximately 10

days, though longer retention times have been observed. Shorter acquisition periods result in shorter retention times. Female whiteflies transmitted TYLCV at rates six times higher than males.

In addition to transmission by whiteflies, TYLCV can also be spread through the movement of infected plant material. This plant material could be in the form of infected tomato transplants or through cuttings taken from infected plants. The virus is not transmitted through seed and not through casual contact with infected plants. The virus can be mechanically transmitted with great effort through the use of special laboratory buffers. Long distance movement could be achieved through the movement of viruliferous whiteflies "hitchhiking" on noninfected plants. It seems likely that TYLCV was imported into the Caribbean, but how that occurred is not known.

A disease cycle for TYLCV in tomato has been proposed in Israel. The virus probably overwinters in the roots of *Cynanchum acutum*, a spring weed which matures in August. A high percent (50%) of *C. acutum* plants sampled were infected with TYLCV. It is believed that whiteflies acquire TYLCV from *C. acutum* plants in early summer and then transmit the virus to tomatoes over a period of several weeks. Whitefly populations are very low in the winter and increase to high populations by summer. Viruliferous whiteflies are first detected in June and their numbers increase rapidly thereafter. Tomatoes are planted in the summer at the same time that whitefly populations are high. The combination of a weed reservoir for TYLCV, high populations of the whitefly vector and young tomato plants creates a devastating situation for tomato production.

MANAGEMENT OF TYLCV IN TOMATO

As the disease cycle discussed above might indicate, TYLCV is a very difficult pathogen to manage. In Israel, Turkey, and Cyprus the virus persists in natural weed reservoirs in the absence of a tomato crop. Whiteflies in many regions readily feed and reproduce on tomato increasing the difficulty of management. A number of different control strategies for TYLCV have been tried with limited success. The most successful strategies to date use a combination of management practices—yellow mulches, timely pesticide applications, and resistant cultivars.

Whitefly Control. As with other plant viruses which can be transmitted for long periods of time by their vectors, suppression of the vector may provide an effective means of reducing virus spread within a field, and reducing the number of vectors which come into the field. Biological control, which often works well in the absence of broad spectrum pesticides to reduce the impact of the whitefly as a pest, offers insufficient control of the vector for the reduction of virus incidence. Insecticidal control is difficult and expensive to maintain throughout an entire cropping season. Although frequent applications of pesticides help to decrease whitefly populations and suppress the spread of TYLCV, virus management through whitefly control is not possible in years where whitefly populations are high. Several foliar-applied insecticides are available for killing whitefly adults and immatures. When applying foliar insecticides, it is essential to maintain good coverage on the underside of the leaves where whiteflies reside. Insecticides with differing modes of action should be rotated to retard selection for whitefly resistance to any one type of pesticide. Soil-applied systemic insecticides have been shown to be effective at reducing spread of TYLCV within tomato fields (secondary spread), but have not been effective at reducing the amount of virus being introduced into fields (primary spread).

Cultural Practices. Yield loss from TYLCV is more severe the earlier the infection begins, therefore nearby infected crops and weeds must be destroyed well in advance of planting. Eradication of overwintering weed hosts in Cyprus significantly reduced the incidence of TYLCV. However, the same practice was not effective in Israel, possibly due to long distance movement of viruliferous whiteflies. In Israel yellow mulches (plastic, straw, sawdust) were found to be effective in decreasing virus spread. (Reflective mulches, which disorient the whiteflies, have been shown to be more effective than colored mulches in reducing the incidence of tomato mottle virus in Florida). The use of trap crops of cucumber, a highly preferred whitefly host, delayed TYLCV spread when planted in alternate rows with, and 30 days before, tomatoes. Fine mesh screens were found to effectively exclude whiteflies and virus from tomato transplant greenhouses. However, this strategy requires extra cooling of the greenhouses due to reduced air flow and may not be economically feasible in Florida. Windbreaks have been tried but resulted in higher whitefly populations and higher numbers of infected plants.

Resistant Cultivars. Fresh market tomato hybrid cultivars are available with tolerance to TYLCV which are adapted to Mediterranean production areas. Plants of the hybrid cultivar TY20 become infected but produce an acceptable yield (22 tons/A compared to 1.6 tons/A for a susceptible cultivar) if plants are protected for four weeks after transplanting using a combination of available control practices.